Exact Finite-Horizon Memory, Conditioning, and Dissipative Decay in Coarse Upwind Finite-Volume Prediction

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Abstract

Coarse finite-volume averages do not generally form a predictive state: realized interface fluxes close a conservative update, but distinct fine-grid states with identical parent averages can generate different future coarse histories. We analyze this failure for periodic scalar advection discretized by a first-order upwind finite-volume method with forward Euler time integration. We derive a finite-horizon observation-rank law: each additional observation exposes one new child-cell layer and contributes one fewer independent direction than the number of parent cells, until the unresolved layers are exhausted. Centralized prediction therefore requires one fewer additional coordinate than the number of parent cells per exposed layer, whereas product-local prediction requires one coordinate per layer in each parent. An anchored flux-divergence queue attains the centralized bound, identifies the periodic flux gauge, and, at saturation, forms a minimal autonomous predictive state with the parent averages. We then distinguish exact observability from stable recoverability. The collar-to-queue map becomes rapidly ill-conditioned as the Courant number decreases, so algebraically visible delayed information may fall below a prescribed numerical tolerance. For Courant numbers strictly between zero and one, we prove contraction of the nonconstant component under bounded arithmetic perturbations, with separate control of mean drift, an explicit perturbation neighborhood, and a grid-dependent decay time. Numerical experiments illustrate the rank ladder, effective-rank loss, queue conditioning, step-function flattening, and delayed coarse separation followed by dissipative decay. The results provide a solvable benchmark for assessing state sufficiency in coarse, reduced, multiscale, and learned scientific models.

Keywords: Finite-volume methods, Coarse graining, Predictive memory, Observability, Numerical dissipation, Stable recovery

MSC Classification: 65M08 , 65F35 , 93B07 , 65M12

1 Introduction

Finite-volume methods evolve cell averages through numerical fluxes at cell interfaces [1–3]. This conservative structure is one reason why finite-volume schemes remain central in computational fluid dynamics (CFD), shock capturing, adaptive mesh refinement, and hyperbolic conservation laws [4–8]. A related question arises whenever a fine calculation is restricted, sampled, reconstructed, or replaced on a coarser grid: do the coarse cell averages alone contain enough information to predict their own future evolution?

In general, the answer is no. A conservative restriction fixes the amount of material in a parent cell but not where that material lies inside the parent. If a parent cell consists of two fine cells, then the fine states

$$(1, 0) \qquad \text{and} \qquad (0, 1)$$

have the same parent average. For right-moving upwind advection, however, the second state has material closer to the right parent interface and can affect the neighboring parent sooner. The parent average is therefore not an exact predictive state.

This observation is simple, but its consequences are easy to obscure. In coarse/fine coupling, a completed fine step can be made conservative on the coarse grid by communicating realized interface fluxes. This is the mechanism behind refluxing and related multilevel corrections [9–11]. Such a realized flux is a one-step message. It does not generally provide all future fluxes, unless additional subcell information or a predictive memory state is retained.

The same distinction appears in reduced modeling and large-eddy simulation. Filtering or averaging produces unresolved stresses, fluxes, or memory terms. Mori–Zwanzig [12, 13] and optimal-prediction viewpoints make this explicit: eliminating variables generally produces non-Markovian resolved dynamics [14–17]. Recent finite-volume closure work also emphasizes that exact unresolved terms can be written down for a declared discretization, even if producing or approximating them remains difficult [18].

The issue has become more visible as machine learning methods have been used for flow super-resolution, spatiotemporal forecasting, and data-driven finite-volume or flux-form prediction [19–30]. In such settings, a coarse input may be compatible with many fine states. Some hidden fine-scale differences may matter for future fluxes, while others may decay because of physical or numerical dissipation. The distinction is essential: exact algebraic information, stable recoverability, and practical finite-precision relevance are different notions.

The present paper isolates these notions in a deliberately simple model: periodic scalar advection with first-order positive upwind forward Euler. The model is simple enough to allow exact theorems, but rich enough to show the main mechanism. Our contribution is not a broad impossibility theorem for turbulence, Euler flows, or learned closures. Instead, it is a precise finite-volume calculation showing what is true in the scalar upwind hierarchy, and why exact rank must be interpreted together with conditioning and dissipation.

1.1 Contributions

Observability, minimal realization, refluxing, unresolved-variable memory, and upwind numerical dissipation are individually classical. The contribution here is their exact finite-volume synthesis for a declared coarse/fine hierarchy: the horizon-by-horizon dimension count, locality-dependent lower bounds, attaining queue coordinates, periodic gauge reduction, and quantitative separation of algebraic visibility from stable and long-time relevance. The paper makes five contributions.

1. We establish the realized-flux identity as the classical one-step baseline: once the integrated interface fluxes of a completed fine step are known, parent averages update exactly by flux telescoping. This explains why refluxing works, but also why a completed flux is not automatically a recursive predictive state.
2. We prove the exact finite-horizon observation-rank formula

   $$\operatorname{rank} \mathcal{O}_L = P + (P-1)\min\{L, r-1\}$$

   for a periodic scalar upwind hierarchy with P parents and r children per parent. This yields continuous centralized and parent-local memory lower bounds.
3. We construct attaining predictive coordinates: raw local queues attain the parent-local bounds, while an anchored flux-divergence queue attains the centralized bound. At saturation, the resulting state is autonomous and has minimal total dimension among continuous autonomous encoders on open support.
4. We quantify stable observability. The queue map for delayed subcell information is triangular with diagonal entries $\lambda, \lambda^2, \ldots, \lambda^q$. Hence exact rank can overstate what is recoverable in finite precision when λ^q approaches the declared singular-value tolerance.
5. We prove dissipative decay under bounded arithmetic perturbations. For $0 < \lambda < 1$, the nonconstant fine-grid component contracts toward an explicit perturbation-controlled neighborhood of the computed mean, while the computed mean satisfies a separate accumulated-drift estimate. The endpoint $\lambda = 1$ is nondissipative.

1.2 Relation to turbulence, multiscale physics, and scientific computing

This issue is especially consequential in turbulence, where coarse representation is not simply a reduction in mesh resolution. Filtering removes nonlinear scale interactions whose influence re-enters the resolved equations through subgrid-scale stresses and, more generally, through history-dependent closure terms. Classical large-eddy simulation (LES) made this dependence explicit through eddy-viscosity closure and its dynamic refinement [31, 32], while projection-operator formulations show that eliminating unresolved degrees of freedom generically produces memory and fluctuating forces [12, 13]. The present analysis does not claim a Navier–Stokes turbulence closure, nor that its scalar rank formula transfers unchanged to turbulent flow. Rather, it provides a fully solvable finite-volume benchmark in which three requirements that are often conflated in turbulence capture are separated exactly: the dimension of the information required for pathwise coarse prediction, the conditioning with which that information can be reconstructed, and the time over which it remains dynamically relevant under dissipation. It thereby provides a precise example of why conservation or high-fidelity reconstruction of coarse observables need not determine the future interface fluxes of the same realization, and why algebraically necessary subgrid information may nevertheless be poorly recoverable or rapidly attenuated at a prescribed tolerance.

The same state-sufficiency question extends beyond classical CFD to scientific-computing methods that advance reduced observables while delegating hidden degrees of freedom to a fine-scale simulator, a reduced representation, or a learned closure. Equation-free and heterogeneous multiscale methods use short microscopic simulations to supply or advance macroscopic information [33, 34]; proper orthogonal decomposition and dynamic mode decomposition seek low-dimensional coordinates for coherent dynamics [35, 36]; and sparse system identification and data-driven discretization infer governing dynamics or numerical operators from observations [22, 37]. In these settings, the exact rank and attaining flux-divergence queue derived here provide a benchmark for state sufficiency, while the singular-value and dissipative estimates impose two additional requirements: the retained state must be stably recoverable and must persist over the intended rollout horizon. The resulting perspective suggests a general validation protocol for reduced, multiscale, and learned models of physical systems: specify the retained observables, prediction horizon, locality or communication architecture, recovery tolerance, and dissipation mechanism before asserting closure, Markovianity, or autonomous prediction.

Active Flux methods provide a complementary finite-volume perspective by evolving interface point values together with cell averages [38, 39]. Their additional interface degrees of freedom are part of a designed high-order discretization. The queue derived here has a different status: it is not proposed as a new discretization, but is the sharp finite-horizon information register for a fixed first-order upwind hierarchy. Moreover, centralized periodic prediction depends only on flux divergences and therefore removes one spatially uniform interface-flux gauge at every queue time.

The scalar upwind problem below is therefore used as a clean reference problem where exact closure dimension, conditioning, and dissipative loss can all be computed. The paper does not claim that the scalar rank formula transfers unchanged to turbulence, nonlinear conservation laws, or general learned closures.

This is useful precisely because it separates questions that are easy to mix:

- exact pathwise closure: what information determines the future coarse trajectory of the same fine realization?
- stable numerical closure: can that information be recovered with reasonable conditioning?
- finite-precision relevance: does unresolved information remain above a declared numerical tolerance or perturbation floor after a dissipative numerical method has acted?

Organization.

The rest of the paper is organized as follows. Section 2 states the one-step finite-volume flux-closure identity and separates completed flux communication from recursive prediction. Section 3 defines the periodic scalar upwind hierarchy used in the analysis. Section 4 proves the exact finite-horizon memory theorem, including centralized and parent-local encoder lower bounds and the flux-divergence queue construction. Section 5 studies stable observability and defines the effective floating-point rank used in the experiments. Section 6 proves dissipative erasure for the nonconstant component of the upwind solution. Section 7 reports the rank, singular-value, conditioning, step-flattening, delayed-collision, and flux-closure experiments. Section 8 interprets the results and Section 8.1 summarizes limitations and future work before the conclusions in Section 9.

2 Finite-volume closure and one-step flux messages

Let a one-dimensional fine grid be partitioned into contiguous parent cells. Fine cell i has width $h_i > 0$, and parent K has total width

$$H_K = \sum_{i \subset K} h_i.$$

The restriction R maps fine cell averages U_i to parent averages:

$$(RU)_K = \frac{1}{H_K} \sum_{i \subset K} h_i U_i.$$

Many fine states have the same parent averages. Exact coarse prediction is therefore a factorization question: can the restricted fine update be written as a function of the declared coarse information?

The first observation is that one completed conservative fine step is always closed by the realized integrated fluxes through parent interfaces.

Proposition 1 (Realized-flux one-step closure)**.** *Suppose one fine finite-volume step has conservative form*

$$U_i^{n+1} = U_i^n - \frac{\Phi_{i+1/2} - \Phi_{i-1/2}}{h_i},$$

where $\Phi_{i+1/2}$ is the realized time-integrated numerical flux through face $i + 1/2$. Let Q_K denote the realized flux through parent interface $K + 1/2$. Then the parent update satisfies

$$(RU^{n+1})_K = (RU^n)_K - \frac{Q_K - Q_{K-1}}{H_K}.$$

Proof Multiply the fine update by h_i and sum over the children of parent K:

$$\sum_{i \subset K} h_i U_i^{n+1} = \sum_{i \subset K} h_i U_i^n - \sum_{i \subset K} (\Phi_{i+1/2} - \Phi_{i-1/2}).$$

All internal fine faces cancel. Only the two exterior parent faces remain, giving

$$\sum_{i \subset K} h_i U_i^{n+1} = \sum_{i \subset K} h_i U_i^n - (Q_K - Q_{K-1}).$$

Division by H_K proves the claim. □

This proposition explains the difference between a one-step conservative message and a recursive predictive state. The realized interface register Q_K can be sufficient after the fine computation has already produced it. It need not determine future realized fluxes. The rest of the paper studies the memory needed for repeated prediction in the scalar upwind hierarchy.

3 Periodic scalar upwind hierarchy

The discrete hierarchy below corresponds to the constant-speed advection equation

$$u_t + a u_x = 0, \qquad a > 0,$$

on a periodic one-dimensional domain. Let h be the fine-cell width, Δt the time step, and

$$\lambda = \frac{a \Delta t}{h}$$

the Courant number. The first-order upwind finite-volume method with forward Euler time integration is

$$u_i^{n+1} = (1 - \lambda) u_i^n + \lambda u_{i-1}^n, \qquad 0 < \lambda \leq 1,$$

with indices understood periodically. In the algebra below we use unit fine-cell widths, which amounts to setting $h = 1$ in the restriction operator.

Let $P \geq 2$ be the number of parent cells and let each parent contain $r \geq 2$ fine cells. The total number of fine cells is

$$N = Pr.$$

Write the fine state as

$$x = (x_{K,j}), \qquad K = 0, \dots, P-1, \qquad j = 0, \dots, r-1.$$

All parent indices are interpreted modulo P, so that $K = -1$ means $K = P - 1$ and $K = P$ means $K = 0$.

The parent average operator $R : \mathbb{R}^{Pr} \to \mathbb{R}^P$ is

$$(Rx)_K = \frac{1}{r} \sum_{j=0}^{r-1} x_{K,j}.$$

Let S be the downstream cyclic shift,

$$(Sx)_{K,j} = \begin{cases} x_{K,j-1}, & j \geq 1, \\ x_{K-1,r-1}, & j = 0. \end{cases}$$

The positive upwind forward-Euler step is

$$A_\lambda = (1 - \lambda)I + \lambda S, \qquad 0 < \lambda \leq 1.$$

For prediction horizon $L \geq 0$, define the observation matrix

$$\mathcal{O}_L = \begin{bmatrix} R \\ RA_\lambda \\ \vdots \\ RA_\lambda^L \end{bmatrix}.$$

The matrix $\mathcal{O}_L$ is a finite-horizon observation matrix. It maps one initial fine-grid state to the sequence of parent averages observed from time 0 through time L. Each block RA_λ^t has size $P \times Pr$, so

$$\mathcal{O}_L \in \mathbb{R}^{(L+1)P \times Pr}.$$

Thus every additional observed time step adds P rows, one for each parent average at that time. The number of rows grows with L, but the rank need not. In the upwind hierarchy studied here, the rank grows only until the exposed horizon reaches $r - 1$ child layers, after which later rows are linear combinations of earlier rows.

The vector $\mathcal{O}_L x$ is the sequence of parent averages generated by the same realized fine state x from time 0 through time L.

We use three notions throughout the paper. First, *exact memory* means the additional real-valued information, beyond the current parent averages, required to reconstruct the parent-average history $\mathcal{O}_L x$ for every fine state in a declared support. Second, *stable observability* refers to whether the algebraically visible directions are visible above a declared singular-value tolerance in floating-point arithmetic. Third, *dissipative erasure* means decay of the nonconstant component of the computed solution, relative to its computed mean, below a prescribed numerical tolerance or perturbation floor. These notions are distinct: exact memory is an algebraic dimension question, stable observability is a conditioning question, and dissipative erasure is a long-time decay question.

4 Exact finite-horizon memory

We first state a standard dimension lower bound. It converts an observation rank into a lower bound on any continuous real-valued encoder. For an open support $\Omega \subset \mathbb{R}^{Pr}$, define the centralized finite-horizon exact excess memory dimension by

$$d_L^{\text{cen}}(\Omega) = \inf \left\{ d \in \mathbb{N}_0 : \begin{array}{l} \text{there exist a continuous } M : \Omega \to \mathbb{R}^d \text{ and a decoder } G \\ \text{such that } \mathcal{O}_L x = G(Rx, M(x)) \text{ for every } x \in \Omega \end{array} \right\}.$$

This quantity counts real-valued coordinates beyond the parent averages. It is not a bit-complexity, communication-cost, or statistical learning quantity.

For a product-local architecture, assume

$$\Omega = \Omega_0 \times \cdots \times \Omega_{P-1}, \qquad \Omega_K \subset \mathbb{R}^r \text{ open.}$$

A vector $m = (m_0, \dots, m_{P-1})$ is called locally admissible at horizon L if there exist continuous parent-local maps

$$M_K : \Omega_K \to \mathbb{R}^{m_K}, \qquad K = 0, \dots, P-1,$$

and a decoder G such that

$$\mathcal{O}_L x = G\big(Rx, M_0(x_0), \dots, M_{P-1}(x_{P-1})\big) \qquad \text{for every } x \in \Omega.$$

The terms encoder and decoder are used here in a finite-dimensional mathematical sense, not in the Shannon-information sense. An encoder is an auxiliary memory map M that extracts additional real-valued coordinates from the fine state, beyond the parent averages. A decoder is a prediction map G that uses the parent averages and this auxiliary memory to reconstruct the desired parent-average history. The lower bounds below concern Euclidean state dimension for continuous maps on open sets; they are not bit-complexity, entropy, mutual-information, or channel-capacity statements. Continuity is required of the encoder; the decoder is only assumed to be defined on the attained encoded data.

The theorem below determines both the centralized dimension and the coordinatewise local lower bound.

Lemma 1 (Continuous encoder lower bound)**.** *Let $H : \mathbb{R}^n \to \mathbb{R}^m$ be linear, and let $\Omega \subset \mathbb{R}^n$ contain a nonempty open set. Let $C : \mathbb{R}^n \to \mathbb{R}^p$ be linear. Suppose there are a continuous map $M : \Omega \to \mathbb{R}^d$ and a decoder G, defined on the attained image, such that*

$$Hx = G(Cx, M(x)), \qquad x \in \Omega.$$

Then

$$\operatorname{rank} C + d \geq \operatorname{rank} H.$$

Proof Let $k = \operatorname{rank} H$ and $c = \operatorname{rank} C$. Replace C by coordinates on its range, so that Cx is viewed as an element of a c-dimensional linear space. Choose a k-dimensional linear complement V of $\ker H$. On a small relative-open ball in an affine copy of V, the map H is injective. If two points in that ball have the same (C, M) value, then the decoder gives them the same H value, and hence the two points are equal. Thus (C, M) continuously injects an open subset of $\mathbb{R}^k$ into $\mathbb{R}^{c+d}$. By invariance of domain, $c + d \geq k$. □

Theorem 1 (Finite-horizon upwind memory)**.** *Let $P \geq 2$, $r \geq 2$, $L \geq 0$, and $0 < \lambda \leq 1$. Put*

$$q = \min\{L, r-1\}.$$

Then

$$\operatorname{rank} \mathcal{O}_L = P + (P-1)q.$$

Consequently, the following encoder lower bounds hold. First, any continuous centralized encoder that reconstructs $\mathcal{O}_L x$ from Rx on an open set requires at least

$$(P-1)q$$

additional real coordinates.

Second, suppose the support has product form

$$\Omega = \Omega_0 \times \cdots \times \Omega_{P-1} \subset (\mathbb{R}^r)^P,$$

where each $\Omega_K \subset \mathbb{R}^r$ is open, and suppose the message is a product of continuous parent-local producers,

$$M(x) = \big(M_0(x_0), \dots, M_{P-1}(x_{P-1})\big), \qquad M_K : \Omega_K \to \mathbb{R}^{m_K},$$

with

$$x_K = (x_{K,0}, \dots, x_{K,r-1}).$$

If a decoder reconstructs $\mathcal{O}_L x$ from Rx and $M(x)$ for all $x \in \Omega$, then

$$m_K \geq q \qquad \textit{for every } K = 0, \dots, P-1.$$

Moreover, the centralized lower bound is attained by the anchored flux-divergence queue

$$\big(JB\Phi_0, \dots, JB\Phi_{q-1}\big) \in \mathbb{R}^{(P-1)q},$$

where

$$\Phi_{K,t} = \lambda(A_\lambda^t x)_{K,r-1}, \qquad (B\Phi_t)_K = \Phi_{K,t} - \Phi_{K-1,t},$$

and J deletes one coordinate of a zero-sum periodic vector.

Proof Since I and S commute,

$$A_\lambda^t = \sum_{j=0}^{t} \binom{t}{j} (1-\lambda)^{t-j} \lambda^j S^j.$$

The change of variables from $(R, RS, \dots, RS^L)$ to $(R, RA_\lambda, \dots, RA_\lambda^L)$ is triangular with nonzero diagonal entries $1, \lambda, \dots, \lambda^L$. Therefore these two families have the same row span.

For $0 \leq t \leq r-2$, define

$$D_t = r(RS^{t+1} - RS^t).$$

Then

$$(D_t x)_K = x_{K-1,r-1-t} - x_{K,r-1-t}.$$

Thus each new shift exposes one new child layer through the periodic difference operator. The P rows of D_t sum to zero, and hence

$$\operatorname{rank} D_t \leq P-1.$$

Moreover, for $t \geq 1$,

$$RS^t = R + \frac{1}{r} \sum_{s=0}^{t-1} D_s.$$

Therefore, for $L \leq r-1$,

$$\operatorname{rank} \operatorname{span}\{R, RS, \dots, RS^L\} \leq P + L(P-1).$$

If $L > r-1$, then S^r shifts by one parent cell, so later rows RS^t, $t \geq r$, lie in the row span of

$$R, RS, \dots, RS^{r-1}.$$

Consequently, for all $L \geq 0$,

$$\operatorname{rank} \operatorname{span}\{R, RS, \dots, RS^L\} \leq P + q(P-1), \qquad q = \min\{L, r-1\}.$$

We now construct a matching independent set of rows.

Let

$$j_\ell = r-1-\ell, \qquad \ell = 0, \dots, q-1,$$

be the q exposed right-collar layers. Since $q \leq r-1$, none of these layers is $j = 0$. For each exposed layer j_ℓ, take the $P-1$ difference rows

$$D_{\ell,K} = r(RS^{\ell+1} - RS^\ell)_K, \qquad K = 1, \dots, P-1,$$

and take the P parent-mean rows R_K, $K = 0, \dots, P-1$. Consider the columns consisting of:

$$(K, j_\ell), \qquad K = 1, \dots, P-1, \quad \ell = 0, \dots, q-1,$$

together with the P columns

$$(K, 0), \qquad K = 0, \dots, P-1.$$

Order the difference rows first, layer by layer, and the parent-mean rows last. Order the columns in the same way, with the child-0 columns last. The resulting square submatrix has block form

$$\begin{bmatrix} D & 0 \\ * & r^{-1} I_P \end{bmatrix}.$$

Here D is block diagonal across the exposed layers. In each layer, the $(P-1) \times (P-1)$ difference block is lower triangular with diagonal entries -1. Hence

$$\det D = \pm 1, \qquad \det \begin{bmatrix} D & 0 \\ * & r^{-1} I_P \end{bmatrix} = \pm r^{-P} \neq 0.$$

Therefore the parent means and the q exposed layer-difference blocks are linearly independent. They give

$$P + (P-1)q$$

independent rows.

No later shift adds rank. Indeed, S^r shifts by one parent cell, so RS^{t+r} is a parent-level cyclic permutation of RS^t. This proves

$$\operatorname{rank} \mathcal{O}_L = P + (P-1)q.$$

The centralized encoder lower bound follows from Lemma 1 with $H = \mathcal{O}_L$, $C = R$, and $p = P$.

We next prove the product-local lower bound. If $q = 0$, there is nothing to prove, so assume $q \geq 1$. Fix a parent K. Choose a point $\bar{x} \in \Omega$. Since Ω_K is open, there is a sufficiently small open ball $B \subset \mathbb{R}^q$ such that the following q-parameter family remains in Ω_K:

$$x_{K,0}(\xi) = \bar{x}_{K,0} - \sum_{\ell=0}^{q-1} \xi_\ell, \qquad x_{K,r-1-\ell}(\xi) = \bar{x}_{K,r-1-\ell} + \xi_\ell, \quad \ell = 0, \ldots, q-1,$$

with all other children of parent K fixed at their values in $\bar{x}$. All other parents are also fixed at their values in $\bar{x}$. The compensation in child 0 keeps the parent average of parent K fixed, so $Rx(\xi)$ is independent of ξ. The messages from all parents other than K are also independent of ξ.

Suppose two parameters $\xi, \xi' \in B$ give the same local message:

$$M_K(x_K(\xi)) = M_K(x_K(\xi')).$$

Then the full data $(Rx, M(x))$ are the same for $x(\xi)$ and $x(\xi')$. Since the decoder reconstructs $\mathcal{O}_L x$, we must have

$$\mathcal{O}_L x(\xi) = \mathcal{O}_L x(\xi').$$

Let

$$d = x(\xi) - x(\xi'), \qquad \delta_\ell = \xi_\ell - \xi'_\ell.$$

The triangular relation between $(R, RS, \ldots, RS^L)$ and $(R, RA_\lambda, \ldots, RA_\lambda^L)$ is invertible. Hence $\mathcal{O}_L d = 0$ implies

$$RS^t d = 0, \qquad 0 \leq t \leq L.$$

For each $\ell = 0, \ldots, q-1$, we have $\ell + 1 \leq L$. Using

$$r(RS^{\ell+1} d - RS^\ell d)_K = d_{K-1,r-1-\ell} - d_{K,r-1-\ell},$$

and noting that only parent K varies, we obtain

$$0 = r(RS^{\ell+1} d - RS^\ell d)_K = -\delta_\ell.$$

Thus $\delta_\ell = 0$ for every ℓ, so $\xi = \xi'$. Therefore the map

$$\xi \mapsto M_K(x_K(\xi))$$

is a continuous injection from an open subset of $\mathbb{R}^q$ into $\mathbb{R}^{m_K}$. By invariance of domain, $m_K \geq q$. Since K was arbitrary, this holds for every parent.

We also record the local raw outflow map, since it gives the attaining local queue and will be used later for conditioning. Let

$$c_{K,j} = x_{K,r-1-j}, \qquad j = 0, \ldots, q-1,$$

be the right-collar values of parent K. Before wraparound, the outgoing flux queue satisfies

$$\Phi_{K,t} = \lambda \sum_{j=0}^{t} \binom{t}{j} (1-\lambda)^{t-j} \lambda^j c_{K,j}, \qquad t = 0, \ldots, q-1.$$

Thus $(\Phi_{K,0}, \ldots, \Phi_{K,q-1})$ is obtained from the collar $(c_{K,0}, \ldots, c_{K,q-1})$ by a lower-triangular matrix with diagonal entries

$$\lambda, \lambda^2, \ldots, \lambda^q.$$

For $0 < \lambda \leq 1$, this map is invertible. Hence the raw q-entry local outflow queue contains exactly the exposed q collar coordinates.

It remains to show attainment. The parent update can be written as

$$RA_\lambda^{t+1} x = RA_\lambda^t x - \frac{1}{r} B\Phi_t.$$

The vector $B\Phi_t$ has zero sum, so deleting one coordinate loses no information. Given Rx and the anchored sequence $(JB\Phi_0, \ldots, JB\Phi_{q-1})$, one reconstructs the full divergence sequence and hence the parent history through the exposed horizon.

If $L \leq r-1$, this is the entire required history. If $L > r-1$, then $q = r-1$, and the same saturated queue reconstructs

$$y_0, y_1, \ldots, y_{r-1}, \qquad y_n = RA_\lambda^n x.$$

The later parent states are then determined by a fixed linear recurrence. Indeed,

$$A_\lambda - (1-\lambda)I = \lambda S,$$

so

$$\big(A_\lambda - (1-\lambda)I\big)^r = \lambda^r S^r.$$

Let Π_P denote the parent-level cyclic shift,

$$(\Pi_P y)_K = y_{K-1}.$$

Multiplying the previous identity by RA_λ^n gives

$$y_{n+r} = \lambda^r \Pi_P y_n - \sum_{j=0}^{r-1} \binom{r}{j} \big(-(1-\lambda)\big)^{r-j} y_{n+j}.$$

Therefore $y_r, y_{r+1}, \dots, y_L$ are obtained recursively from $y_0, \dots, y_{r-1}$. Hence the saturated anchored queue determines $\mathcal{O}_L x$ for every $L \geq r-1$. □

In the notation above, Theorem 1 gives

$$d_L^{\mathrm{cen}}(\Omega) = (P-1)\min\{L, r-1\}$$

for every open support Ω on which the stated encoder problem is posed. For product-local encoders, every admissible vector $(m_0, \dots, m_{P-1})$ satisfies

$$m_K \geq \min\{L, r-1\}, \qquad K = 0, \dots, P-1,$$

and the raw local outflow queues attain this coordinatewise bound.

Corollary 1 (All-horizon invisible subspace)**.** *For $L \geq r-1$, the nullspace of the saturated observation operator is*

$$\mathcal{N}_\infty = \left\{ x \in \mathbb{R}^{Pr} : x_{K,j} = c_j \text{ for every } K, \quad \sum_{j=0}^{r-1} c_j = 0 \right\}.$$

Consequently,

$$\dim \mathcal{N}_\infty = r-1.$$

These are precisely the fine-grid directions that remain invisible to all future parent-average observations.

Proof If $x_{K,j} = c_j$ for every parent K, then every parent contains the same child profile. Since $\sum_{j=0}^{r-1} c_j = 0$, all parent averages are zero. Since A_λ is a polynomial in the periodic shift S, each future state is a linear combination of shifted copies of the same repeated zero-mean profile. Hence all future parent averages remain zero.

Conversely, for $L \geq r-1$, Theorem 1 gives

$$\operatorname{rank} \mathcal{O}_L = P + (P-1)(r-1) = Pr - r + 1.$$

Thus the nullity is

$$Pr - (Pr - r + 1) = r - 1.$$

The displayed invisible subspace has dimension $r-1$, because it is the space of r-component profiles with zero sum. Therefore it is the entire saturated nullspace. □

Corollary 2 (Sharp finite delay)**.** *For every $m \in \{0, \dots, r-2\}$, there exist two fine-grid states whose parent averages agree from time 0 through time m, but differ at time $m+1$. Conversely, if two fine-grid states have identical parent-average histories through time $r-1$, then their parent-average histories agree for all subsequent times.*

Proof Fix a parent K, choose $\delta \neq 0$, and set

$$\ell = r - 1 - m.$$

Consider the difference vector

$$v = \delta(e_{K,0} - e_{K,\ell}).$$

The two nonzero entries of v lie in the same parent and have zero parent average. For $0 \le t \le m$, every shift $S^s v$ appearing in $A_\lambda^t v$, with $0 \le s \le t$, still has both nonzero entries inside the same parent. Hence

$$RA_\lambda^t v = 0, \qquad 0 \le t \le m.$$

At time $m+1$, the component initially at child ℓ reaches the next parent. The term with S^{m+1} has coefficient λ^{m+1}, so

$$RA_\lambda^{m+1} v = \frac{\lambda^{m+1}\delta}{r}(e_K - e_{K+1}) \neq 0,$$

with parent indices interpreted periodically. This proves the sharp delayed separation.

The converse follows from the recurrence in the proof of Theorem 1. Once the parent-average history through time $r-1$ is fixed, the saturated recurrence determines every later parent average. □

Corollary 3 (Saturated memory and periodic gauge)**.** *At saturation, $q = r-1$. A product-local upstream encoder needs $r-1$ coordinates per parent: each parent must retain its unresolved downstream, or outflow, collar. A centralized encoder needs only*

$$(P-1)(r-1)$$

coordinates beyond the parent means. The saving of $r-1$ coordinates is exactly the spatially uniform periodic flux gauge, one redundant constant face-flux mode at each queue time.

Proof Set $q = r-1$ in Theorem 1. The product-local lower bound gives $m_K \ge r-1$ for every parent K. The local raw outflow queue has exactly $r-1$ entries and, by the triangular collar-to-queue map in the proof of Theorem 1, it recovers the $r-1$ right-collar values of the parent. The remaining child value is then determined by the parent average.

For the centralized encoder, the anchored divergence queue has $(P-1)(r-1)$ coordinates. At each queue time, adding the same constant face flux to every periodic parent interface produces no change in the parent-average update, since only cyclic flux differences enter the update. This removes exactly one scalar gauge coordinate per queue time, hence exactly $r-1$ coordinates at saturation. □

Corollary 4 (Autonomous saturated predictive state)**.** *Let*

$$y_n = RA_\lambda^n x$$

and define the saturated state

$$Z_n = (y_n, JB\Phi_n, JB\Phi_{n+1}, \ldots, JB\Phi_{n+r-2}).$$

Then there exists a fixed linear map $\mathcal{F}$, depending only on (P, r, λ), such that

$$Z_{n+1} = \mathcal{F} Z_n$$

for every $n \ge 0$. Thus the parent means together with the saturated flux-divergence queue form an autonomous predictive state and advance without revisiting the fine-grid state.

Proof Given Z_n, the anchored divergence entries reconstruct

$$B\Phi_n, \ldots, B\Phi_{n+r-2},$$

because each $B\Phi_t$ has zero sum. Starting from y_n, the update

$$y_{t+1} = y_t - \frac{1}{r} B\Phi_t$$

then reconstructs

$$y_{n+1}, \ldots, y_{n+r-1}.$$

The recurrence from the proof of Theorem 1,

$$y_{n+r} = \lambda^r \Pi_P y_n - \sum_{j=0}^{r-1} \binom{r}{j} \bigl(-(1-\lambda)\bigr)^{r-j} y_{n+j},$$

gives y_{n+r}. Finally,

$$B\Phi_{n+r-1} = r(y_{n+r-1} - y_{n+r})$$

provides the new tail of the queue. Every operation is linear and depends only on (P, r, λ), so the update has the form $Z_{n+1} = \mathcal{F} Z_n$. □

Corollary 5 (Minimal autonomous predictive-state dimension). *Let $\Omega \subset \mathbb{R}^{Pr}$ contain a nonempty open set. Let $E : \Omega \to \mathbb{R}^d$ be a continuous encoder. Suppose there exist maps*

$$F : \mathbb{R}^d \to \mathbb{R}^d, \qquad H_{\text{out}} : \mathbb{R}^d \to \mathbb{R}^P,$$

such that

$$RA_\lambda^n x = H_{\text{out}}\big(F^n(E(x))\big) \qquad \textit{for every } x \in \Omega \textit{ and every } n \geq 0.$$

Then

$$d \geq \operatorname{rank} \mathcal{O}_{r-1} = Pr - r + 1.$$

The saturated state in Corollary 4 attains this dimension.

Proof The assumed autonomous representation determines

$$RA_\lambda^n x, \qquad n = 0, \ldots, r-1,$$

from $E(x)$. Hence $\mathcal{O}_{r-1}x$ factors through the continuous map $E : \Omega \to \mathbb{R}^d$. Applying Lemma 1 with $H = \mathcal{O}_{r-1}$ and $C = 0$ gives

$$d \geq \operatorname{rank} \mathcal{O}_{r-1}.$$

By Theorem 1,

$$\operatorname{rank} \mathcal{O}_{r-1} = P + (P-1)(r-1) = Pr - r + 1.$$

The state in Corollary 4 has exactly $P + (P-1)(r-1) = Pr - r + 1$ coordinates, so the bound is sharp. $\square$

5 Stable observability and effective rank

The exact rank theorem is algebraic. It says which directions are visible in exact arithmetic. It does not say whether those directions are stably recoverable from floating-point data. This distinction is standard in numerical linear algebra [40, 41] and is closely related to observability questions for discretized partial differential equations (PDEs) [42]. Classical realization and model-reduction theory also connect finite input-output histories, observability, singular values, and minimal state representations [43, 44].

Let

$$c_j = x_{K,r-1-j}$$

denote the right collar values of one parent. Before wraparound, the outgoing flux queue satisfies

$$\Phi_t = \lambda \sum_{j=0}^{t} \binom{t}{j} (1-\lambda)^{t-j} \lambda^j c_j, \qquad t = 0, \ldots, q-1.$$

This defines a lower-triangular matrix T_q mapping the q collar values to the first q outflow values.

Theorem 2 (Conditioning of delayed queue recovery). *For $q \geq 1$, let T_q be the collar-to-queue matrix defined by*

$$(T_q)_{tj} = \binom{t}{j} (1-\lambda)^{t-j} \lambda^{j+1}, \qquad 0 \leq j \leq t \leq q-1,$$

and $(T_q)_{tj} = 0$ for $j > t$. Then T_q is lower triangular with diagonal entries

$$\lambda, \lambda^2, \ldots, \lambda^q,$$

and

$$\det T_q = \lambda^{q(q+1)/2}.$$

Its inverse is also lower triangular and is given by

$$(T_q^{-1})_{tj} = \lambda^{-(t+1)} \binom{t}{j} \big[-(1-\lambda)\big]^{t-j}, \qquad 0 \leq j \leq t \leq q-1.$$

Moreover,

$$\|T_q\|_\infty = \lambda, \qquad \|T_q^{-1}\|_\infty = \lambda^{-q}(2-\lambda)^{q-1},$$

so

$$\kappa_\infty(T_q) = \left(\frac{2-\lambda}{\lambda}\right)^{q-1}.$$

Finally,

$$\frac{\lambda^q}{\sqrt{q}\,(2-\lambda)^{q-1}} \le \sigma_{\min}(T_q) \le \lambda^q.$$

Thus, for fixed q, the smallest singular value scales like λ^q as $\lambda \to 0$.

Proof The triangular form and diagonal entries follow immediately from the definition of T_q. The determinant is the product of the diagonal entries.

The stated inverse follows from the binomial inversion identity. Direct multiplication gives, for $j \le t$,

$$\sum_{m=j}^{t} \binom{t}{m} (1-\lambda)^{t-m} \lambda^{m+1} \lambda^{-(m+1)} \binom{m}{j} \left[-(1-\lambda)\right]^{m-j} = \binom{t}{j} (1-\lambda)^{t-j} \sum_{m=j}^{t} \binom{t-j}{m-j} (-1)^{m-j}.$$

The final sum is 1 when $t = j$ and 0 otherwise, proving the inverse formula.

For the infinity norm,

$$\sum_{j=0}^{t} |(T_q)_{tj}| = \lambda \sum_{j=0}^{t} \binom{t}{j} (1-\lambda)^{t-j} \lambda^j = \lambda.$$

Similarly,

$$\sum_{j=0}^{t} |(T_q^{-1})_{tj}| = \lambda^{-(t+1)} \sum_{j=0}^{t} \binom{t}{j} (1-\lambda)^{t-j} = \lambda^{-(t+1)} (2-\lambda)^t,$$

whose maximum over $t = 0, \ldots, q-1$ is $\lambda^{-q}(2-\lambda)^{q-1}$. This proves the formula for $\kappa_\infty(T_q)$.

The upper bound on $\sigma_{\min}(T_q)$ follows by applying T_q to the last coordinate vector:

$$\sigma_{\min}(T_q) \le \|T_q e_{q-1}\|_2 = \lambda^q.$$

For the lower bound, use

$$\sigma_{\min}(T_q) = \|T_q^{-1}\|_2^{-1}$$

and

$$\|T_q^{-1}\|_2 \le \sqrt{q}\, \|T_q^{-1}\|_\infty.$$

This gives the stated two-sided estimate. □

Proposition 2 (Spatial conditioning of the periodic divergence)**.** *Let $B : \mathbb{R}^P \to \mathbb{R}^P$ be the periodic difference operator*

$$(B\phi)_K = \phi_K - \phi_{K-1}.$$

Then $B\mathbf{1} = 0$, and the nonzero singular values of B are

$$\sigma_k(B) = 2\left|\sin\left(\frac{\pi k}{P}\right)\right|, \qquad k = 1, \ldots, P-1.$$

In particular,

$$\sigma_{\min}^{+}(B) = 2\sin\left(\frac{\pi}{P}\right),$$

where $\sigma_{\min}^{+}$ denotes the smallest nonzero singular value. Thus centralized recovery through flux differences has both a temporal conditioning mechanism, represented by T_q, and a spatial conditioning mechanism, represented by B.

Proof The operator B is circulant. Its Fourier eigenvalues are

$$1 - e^{-2\pi i k/P}, \qquad k = 0, \ldots, P-1.$$

The corresponding singular values are their moduli,

$$|1 - e^{-2\pi i k/P}| = 2\left|\sin\left(\frac{\pi k}{P}\right)\right|.$$

The $k = 0$ mode is the constant vector and gives the periodic flux gauge. The smallest nonzero value occurs at $k = 1$ and $k = P-1$. □

For algebraic dimension counting, deleting one coordinate of a zero-sum periodic vector is sufficient. For conditioning analysis, an orthonormal basis of $\mathbf{1}^\perp$ gives a symmetric representation of the same gauge quotient. The experiments use the algebraic quotient, while Proposition 2 records the intrinsic conditioning of the periodic difference itself.

We use exact equalities for exact arithmetic and a separate tolerance model for floating-point arithmetic. In the numerical experiments we use a tolerance-dependent effective rank computed from a singular value decomposition (SVD). Since

$$\mathcal{O}_L \in \mathbb{R}^{(L+1)P\times Pr},$$

and since $s_i(\mathcal{O}_L)$ denotes the singular values of $\mathcal{O}_L$, counted with multiplicity, we define

$$\operatorname{rank}_\tau(\mathcal{O}_L) = \operatorname{card}\{i : s_i(\mathcal{O}_L) > \tau\}, \qquad \tau = C_{\mathrm{svd}}\epsilon_{\mathrm{mach}} \max\{(L+1)P, Pr\}\|\mathcal{O}_L\|_2.$$

Here $C_{\mathrm{svd}} = 100$ in all reported computations. The factor $\max\{(L+1)P, Pr\}$ follows the usual matrix-size scaling used in singular-value based numerical-rank decisions [40, 41]. This definition is not a new mathematical rank; it is a declared floating-point diagnostic.

This is the appropriate way to account for machine precision. Adding an informal "$+\epsilon_{\mathrm{mach}}$" to every identity would obscure the difference between exact algebra and numerical tolerance.

6 Dissipative erasure

For $0 < \lambda < 1$, first-order upwind is dissipative. This changes the practical meaning of missing information. A hidden subcell difference may be required for exact short-time closure, yet later be damped below a declared perturbation floor.

The result in this section is a bounded-perturbation theorem. It does not derive a complete implementation-specific Institute of Electrical and Electronics Engineers (IEEE) floating-point roundoff model. Instead, it assumes that the computed update is the exact upwind update plus a per-step perturbation with separately bounded mean and nonconstant components. This is the level of modeling used in the numerical interpretation below.

At the level of modified-equation analysis [45–48], the same dissipative mechanism appears as an artificial diffusion term. For

$$\lambda = \frac{a\Delta t}{h},$$

the first-order upwind update has the formal modified equation

$$u_t + au_x = \frac{ah}{2}(1-\lambda)u_{xx} + \mathcal{O}(h^2).$$

Thus the contraction per update and the numerical diffusion per unit physical time are related but distinct quantities. This also explains why the control case $\lambda = 1$ is nondissipative: the leading artificial diffusion coefficient vanishes.

Let P_0 denote the projection onto the spatial mean,

$$P_0 z = \bar{z}\mathbf{1}, \qquad \bar{z} = \frac{1}{N}\sum_{i=0}^{N-1} z_i.$$

In this section $z^n \in \mathbb{R}^N$ denotes the evolving fine-grid solution vector; it is the same type of object denoted by u^n in the numerical experiments.

Theorem 3 (Dissipative decay under bounded arithmetic perturbations)**.** *Let $0 < \lambda < 1$, and suppose the computed solution satisfies*

$$\tilde{z}^{n+1} = A_\lambda \tilde{z}^n + \delta^n,$$

with

$$\|(I-P_0)\delta^n\|_2 \le \eta, \qquad |\overline{\delta^n}| \le \eta_m.$$

Here

$$\overline{\delta^n} = \frac{1}{N}\mathbf{1}^\top \delta^n$$

is the spatial mean of the arithmetic perturbation at step n.

Define

$$\rho_N = \sqrt{1 - 4\lambda(1-\lambda)\sin^2\left(\frac{\pi}{N}\right)}.$$

Then $0 < \rho_N < 1$, and

$$\|(I - P_0)\tilde{z}^n\|_2 \le \rho_N^n \|(I - P_0)\tilde{z}^0\|_2 + \frac{1-\rho_N^n}{1-\rho_N}\eta.$$

The computed mean satisfies

$$|\bar{\tilde{z}}^n - \bar{\tilde{z}}^0| \le n\eta_m.$$

Finally,

$$\|R\tilde{z}^n - \bar{\tilde{z}}^n \mathbf{1}_P\|_2 \le \frac{1}{\sqrt{r}}\left[\rho_N^n \|(I - P_0)\tilde{z}^0\|_2 + \frac{1-\rho_N^n}{1-\rho_N}\eta\right].$$

Proof First note the elementary dissipation identity. Since the cyclic shift S is orthogonal,

$$\|A_\lambda z\|_2^2 = \|z\|_2^2 - \lambda(1-\lambda)\|(I - S)z\|_2^2.$$

Thus the scheme is nondissipative at $\lambda = 1$, while for $0 < \lambda < 1$ every nonconstant component loses discrete L^2 energy. The Fourier argument below gives the sharp contraction rate on the zero-mean subspace.

The matrix A_λ is circulant and normal. Its Fourier eigenvalues are

$$\mu_k = (1-\lambda) + \lambda e^{-2\pi i k/N}, \qquad k = 0, \ldots, N-1.$$

For $k \ne 0$,

$$|\mu_k|^2 = 1 - 4\lambda(1-\lambda)\sin^2\left(\frac{\pi k}{N}\right) \le \rho_N^2.$$

Hence

$$\|A_\lambda v\|_2 \le \rho_N \|v\|_2 \qquad \text{whenever } P_0 v = 0.$$

Set $v^n = (I - P_0)\tilde{z}^n$. Then

$$\|v^{n+1}\|_2 \le \rho_N \|v^n\|_2 + \eta.$$

Iteration gives the first estimate. Averaging the perturbation equation gives the mean-drift estimate. The restriction estimate follows from Cauchy's inequality:

$$\|Rv\|_2 \le r^{-1/2}\|v\|_2.$$

□

Corollary 6 (Step-function flattening)**.** *Let*

$$z_i^0 = \begin{cases} 1, & 0 \le i < m, \\ 0, & m \le i < N, \end{cases} \qquad \alpha = \frac{m}{N}.$$

Then

$$\|z^0 - \alpha\mathbf{1}\|_2 = \sqrt{N\alpha(1-\alpha)}.$$

Under the bounded-perturbation model of Theorem 3,

$$\max_i |\tilde{z}_i^n - \alpha| \le \rho_N^n \sqrt{N\alpha(1-\alpha)} + \frac{1-\rho_N^n}{1-\rho_N}\eta + n\eta_m.$$

Thus the periodic upwind solution flattens about its computed mean up to a declared perturbation floor. It remains close to the initial mean α only to the extent quantified by the mean-drift term $n\eta_m$.

Proof The mean of the initial step is

$$\alpha = \frac{1}{N}\sum_{i=0}^{N-1} z_i^0 = \frac{m}{N}.$$

Hence

$$\|z^0 - \alpha\mathbf{1}\|_2^2 = m(1-\alpha)^2 + (N-m)\alpha^2.$$

Using $m = N\alpha$, this becomes

$$\|z^0 - \alpha\mathbf{1}\|_2^2 = N\alpha(1-\alpha)^2 + N(1-\alpha)\alpha^2 = N\alpha(1-\alpha).$$

Therefore

$$\|z^0 - \alpha \mathbf{1}\|_2 = \sqrt{N\alpha(1-\alpha)}.$$

By Theorem 3,

$$\|\tilde{z}^n - \bar{\tilde{z}}^n \mathbf{1}\|_2 \le \rho_N^n \sqrt{N\alpha(1-\alpha)} + \frac{1-\rho_N^n}{1-\rho_N}\eta.$$

Moreover,

$$|\bar{\tilde{z}}^n - \alpha| = |\bar{\tilde{z}}^n - \bar{\tilde{z}}^0| \le n\eta_m,$$

assuming the computed initial state has mean $\bar{\tilde{z}}^0 = \alpha$. For each component,

$$|\tilde{z}_i^n - \alpha| \le |\tilde{z}_i^n - \bar{\tilde{z}}^n| + |\bar{\tilde{z}}^n - \alpha|.$$

Taking the maximum over i, and using

$$\|\tilde{z}^n - \bar{\tilde{z}}^n \mathbf{1}\|_\infty \le \|\tilde{z}^n - \bar{\tilde{z}}^n \mathbf{1}\|_2,$$

gives the stated estimate. □

Remark 1 (Erasure time and grid dependence)**.** *Let*

$$E_0 = \|(I - P_0)\tilde{z}^0\|_2, \qquad E_{\text{floor}} = \frac{\eta}{1-\rho_N}.$$

If a tolerance $\zeta > E_{\text{floor}}$ is prescribed, then the estimate in Theorem 3 guarantees

$$\|(I - P_0)\tilde{z}^n\|_2 \le \zeta$$

whenever

$$n \ge \frac{\log\left(E_0/(\zeta - E_{\text{floor}})\right)}{-\log \rho_N}.$$

Since

$$1 - \rho_N \sim \frac{2\pi^2\lambda(1-\lambda)}{N^2} \qquad \text{as } N \to \infty,$$

the erasure step count satisfies

$$n_{\text{erase}} = O\left(\frac{N^2}{\lambda(1-\lambda)} \log \frac{E_0}{\zeta - E_{\text{floor}}}\right),$$

provided that $\zeta > E_{\text{floor}}$. If E_0 and $\zeta - E_{\text{floor}}$ remain bounded independently of N, this reduces to the fixed-tolerance $N^2/[\lambda(1-\lambda)]$ scaling. If the floor E_{floor} increases with N, the condition $\zeta > E_{\text{floor}}$ may eventually fail for a fixed tolerance.

The corresponding physical-time scaling is also grid dependent. On a fixed periodic domain of length L_{dom}, with

$$h = \frac{L_{\text{dom}}}{N}, \qquad \Delta t = \frac{\lambda h}{a} = \frac{\lambda L_{\text{dom}}}{aN},$$

one obtains

$$t_{\text{erase}} = n_{\text{erase}}\Delta t = O\left(\frac{L_{\text{dom}} N}{a(1-\lambda)} \log \frac{E_0}{\zeta - E_{\text{floor}}}\right).$$

Therefore, dissipative erasure is a long-time finite-grid effect, and the grid dependence must be stated explicitly.

Remark 2. *The endpoint $\lambda = 1$ is excluded from Theorem 3. In that case $A_1 = S$ is a unitary cyclic shift. A step function translates without numerical flattening.*

7 Numerical experiments

The numerical experiments separate five effects and one consistency check. The first experiment checks the exact rank ladder and its tolerance-dependent effective rank. The second displays the singular spectrum at the saturated horizon, showing which exact directions are close to the floating-point threshold. The third tests the conditioning of the collar-to-queue map. The fourth and fifth finite-volume simulations show dissipative flattening of a step function and a delayed same-parent-mean collision followed by erasure. A final scalar flux check verifies the one-step identity in Proposition 1. All computations use double precision, and the SVD tolerance is declared explicitly.

7.1 Common computational setup

All experiments use the scalar periodic upwind update

$$u_i^{n+1} = (1-\lambda)u_i^n + \lambda u_{i-1}^n, \qquad i = 0, \dots, N-1,$$

with periodic indexing. The fine grid has

$$N = Pr$$

cells, where P is the number of parent cells and r is the number of fine cells per parent. Fine cell $i = Kr + j$ belongs to parent K, with $j = 0, \dots, r-1$.

The parent-average restriction matrix $R \in \mathbb{R}^{P \times Pr}$ is defined by

$$R_{K,i} = \begin{cases} 1/r, & Kr \le i \le Kr + r - 1, \\ 0, & \text{otherwise.} \end{cases}$$

The cyclic downstream shift matrix $S \in \mathbb{R}^{N \times N}$ is defined by

$$(Su)_i = u_{i-1},$$

with periodic indexing. The one-step upwind matrix is

$$A_\lambda = (1-\lambda)I + \lambda S.$$

For a horizon L, the observation matrix used in the rank experiments is

$$\mathcal{O}_L = \begin{bmatrix} R \\ RA_\lambda \\ \vdots \\ RA_\lambda^L \end{bmatrix} \in \mathbb{R}^{(L+1)P \times Pr}.$$

Thus $\mathcal{O}_L x$ is the parent-average history generated by the fine state x.

All reported matrix ranks are computed in double precision. The effective rank is the SVD diagnostic defined in Section 5. It is tolerance-dependent and should not be confused with the exact algebraic rank. In the time-dependent finite-volume experiments, the horizontal axis is the time-step index n. Since

$$\Delta t = \frac{\lambda h}{a},$$

equal numbers of time steps do not represent equal physical times for different λ. When $h = 1$ and $a = 1$, the physical time is $t_n = n\lambda$, and the number of full-domain advective periods is $n\lambda/N$.

7.2 Experiment 1: exact rank versus effective rank

Take $P = 8$, $r = 6$, and

$$\lambda \in \{1, 0.5, 0.1, 0.01, 0.001\}.$$

For $L = 0, \dots, r+3$, form

$$\mathcal{O}_L = \begin{bmatrix} R \\ RA_\lambda \\ \vdots \\ RA_\lambda^L \end{bmatrix}.$$

The exact theorem predicts the rank sequence

$$8,\ 15,\ 22,\ 29,\ 36,\ 43,\ 43, \dots.$$

The effective numerical rank is computed using the definition in Section 5, namely

$$\operatorname{rank}_\tau(\mathcal{O}_L) = \operatorname{card}\{i : s_i(\mathcal{O}_L) > \tau\}, \qquad \tau = 100\epsilon_{\text{mach}} \max\{(L+1)P, Pr\} \|\mathcal{O}_L\|_2.$$

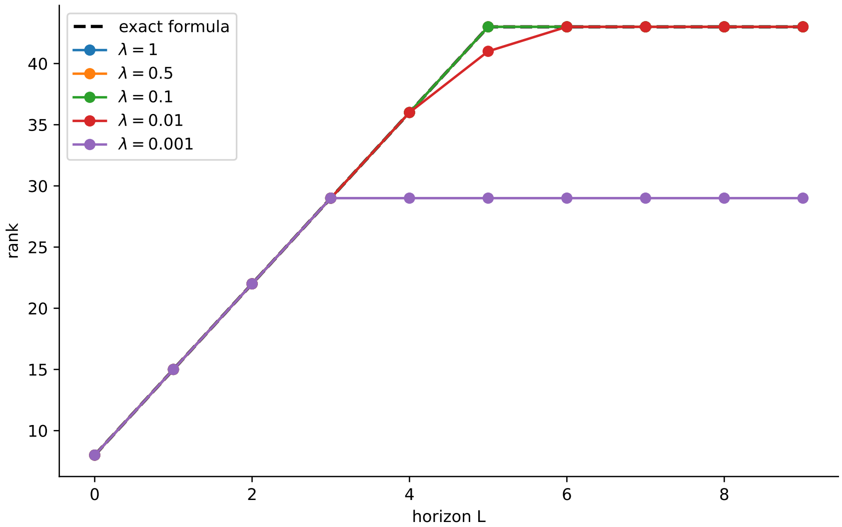

Fig. 1: Exact rank and effective floating-point rank of the observation matrix $\mathcal{O}_L$ for $P = 8$, $r = 6$, and $L = 0, \ldots, 9$. The black dashed curve is the exact formula $P + (P - 1)\min\{L, r - 1\}$, giving the sequence $8, 15, 22, 29, 36, 43, 43, \ldots$. The colored curves show the effective numerical rank computed from the singular values using the tolerance $\tau = 100\epsilon_{\text{mach}} \max\{(L+1)P, Pr\}\|\mathcal{O}_L\|_2$. Several colored curves are not separately visible because their effective ranks coincide with one another, and in some cases also coincide with the black dashed exact-rank curve. This overlap is expected and indicates that the declared SVD tolerance detects the full algebraic rank at those horizons. Visible separation from the dashed curve indicates effective-rank loss.

The diagnostic point of Figure 1 is not that the theorem fails. The exact rank is independent of floating-point tolerance for every $\lambda > 0$. The overlapping curves should be read as agreement, not as missing data: for those horizons, the effective rank is identical for several Courant numbers and often coincides with the exact-rank formula. The visible departures from the dashed curve show where exact rank and stable observable rank differ. Small λ makes the deepest delayed directions nearly invisible under the declared SVD tolerance.

One feature of Figure 1 is that the effective rank may increase after the exact rank has already saturated. This is possible because later rows of $\mathcal{O}_L$ do not add new algebraic directions, but they can strengthen already-visible directions. Equivalently, the finite-horizon observability Gramian

$$W_L = \mathcal{O}_L^\top \mathcal{O}_L = \sum_{t=0}^{L} (A_\lambda^t)^\top R^\top R A_\lambda^t$$

satisfies

$$W_{L+1} = W_L + (A_\lambda^{L+1})^\top R^\top R A_\lambda^{L+1} \succeq W_L.$$

Thus no observed direction becomes weaker when another observation time is added, although the algebraic row space may already be saturated. Longer observation can therefore improve stable recoverability without changing the exact rank.

7.3 Experiment 2: saturated singular values

The rank plot in Figure 1 compresses singular-value information into a single integer. To see where the effective-rank loss comes from, we fix the horizon at the saturated value $L = r - 1 = 5$ and plot the singular values of the single matrix $\mathcal{O}_{r-1}$ (Figure 2). Thus the horizontal axis in Figure 2 is not time or horizon; it is the singular-value index. For the present parameters,

$$\mathcal{O}_{r-1} \in \mathbb{R}^{48\times 48}, \qquad \operatorname{rank} \mathcal{O}_{r-1} = 43,$$

so exactly five singular values vanish in exact arithmetic. The effective rank is the number of singular values that lie above the declared SVD tolerance.

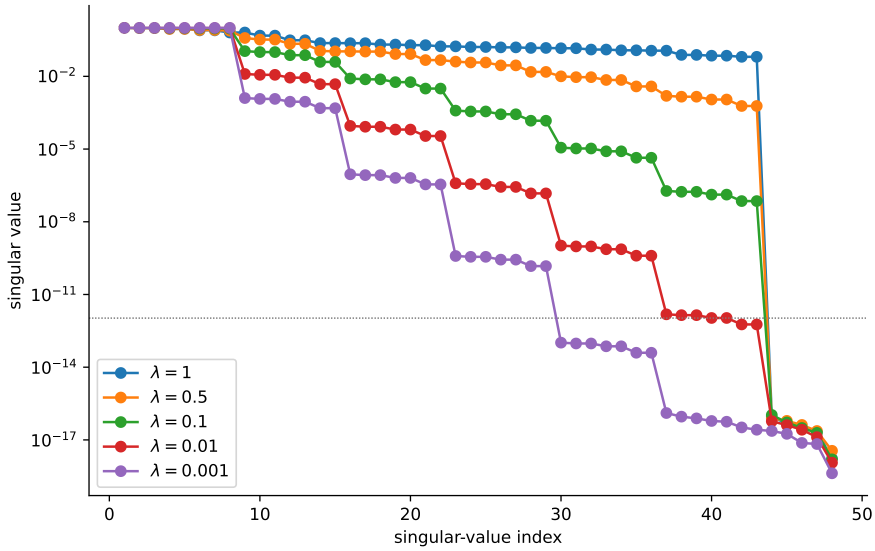

Fig. 2: Singular values of the saturated observation matrix $\mathcal{O}_{r-1}$ for $P = 8$, $r = 6$, and $L = r - 1 = 5$. Unlike Figure 1, this figure does not plot rank versus horizon. The horizon is fixed, and the horizontal axis is the singular-value index $i = 1, \ldots, 48$. Each curve shows the singular values $s_i(\mathcal{O}_{r-1})$ in decreasing order for one value of λ. The exact saturated rank is 43, so five singular values are zero in exact arithmetic. The effective rank reported in Table 1 is obtained by counting how many singular values lie above the declared tolerance. For small λ, some algebraically nonzero singular values fall below this tolerance, which explains the effective-rank loss seen in Figure 1. The dotted horizontal line marks the corresponding SVD tolerance; for these parameters the tolerance levels nearly coincide across λ.

Table 1: Effective rank at the saturated horizon $L = r - 1 = 5$ for $P = 8$, $r = 6$. The exact rank is 43. The effective rank uses the same SVD tolerance as Figure 1. The following table quantifies how small λ can make algebraically visible directions numerically invisible.

λ	Exact rank	Effective rank	Tolerance τ
1	43	43	1.066×10^{-12}
0.5	43	43	1.066×10^{-12}
0.1	43	43	1.066×10^{-12}
0.01	43	41	1.066×10^{-12}
0.001	43	29	1.066×10^{-12}

Table 1 gives the numerical consequence of the singular-value threshold at the saturated horizon. For $\lambda = 1$, 0.5, and 0.1, the effective rank equals the exact rank 43. For $\lambda = 0.01$, two algebraically nonzero directions fall below tolerance, and for $\lambda = 0.001$ the effective rank drops to 29. This is not a contradiction of Theorem 1. It shows that the exactly visible delayed layers can be numerically unrecoverable when their singular values are near the floating-point threshold.

The multiplier $C_{\mathrm{svd}} = 100$ in the threshold is a declared numerical diagnostic, not a mathematical constant. Changing it can change the reported effective rank when singular values lie near the threshold. For this reason, Figure 2 is included alongside the rank table: it exposes the singular spectrum directly, so that other reasonable thresholds can be assessed from the same data.

7.4 Experiment 3: queue conditioning

For $q = 1, \dots, r-1$, construct the triangular queue map T_q from Theorem 2. Compute

$$\sigma_{\min}(T_q) \qquad \text{and} \qquad \lambda^q$$

over a logarithmic grid of λ values. The figure displays the smallest singular value and the reference scaling λ^q. The condition-number behavior is given analytically in Theorem 2.

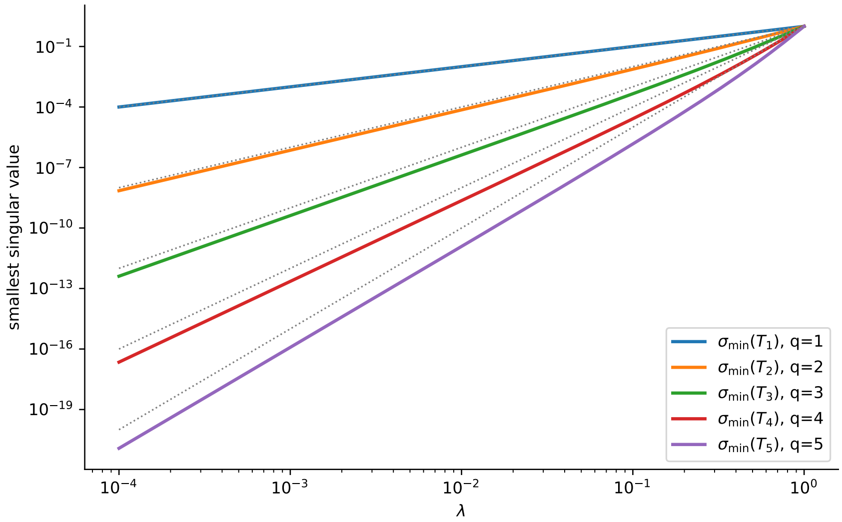

Fig. 3: Conditioning of the collar-to-queue map T_q from Theorem 2. Each solid curve shows $\sigma_{\min}(T_q)$ as a function of λ for $q = 1, \dots, 5$. The dotted reference curves show λ^q. The deepest exposed collar coordinate enters the queue with a factor of order λ^q, so exact recovery becomes ill-conditioned when λ is small. This explains why exact observability and floating-point effective observability can differ.

Figure 3 gives a local explanation of the global SVD behavior. The curves follow the powers of λ predicted by Theorem 2: the deepest exposed collar coordinate enters the queue with a coefficient of order λ^q. Thus the observation matrix can have full exact rank while the information needed to recover a deep collar layer is amplified by a large inverse factor. The figure supports the distinction between exact observability and stable finite-precision observability. For very small λ, some displayed singular values lie below the level at which double precision can be expected to provide high relative accuracy for the smallest singular value. Those points are included only as a visual diagnostic of the trend, not as high-relative-accuracy numerical measurements. The conclusion does not rely on those tiny computed values: the scaling and conditioning are established analytically in Theorem 2.

7.5 Experiment 4: periodic step-function flattening

Use $N = 64$ fine cells and the periodic step

$$u_i^0 = \begin{cases} 1, & 0 \le i < N/2, \\ 0, & N/2 \le i < N. \end{cases}$$

In this experiment, we write the computed fine-grid solution as u^n.

Run the scalar upwind update for $\lambda = 0.5$, $\lambda = 0.1$, and the control case $\lambda = 1$. The comparison is made per update step n. Since $\Delta t = \lambda h/a$, the three curves do not correspond to equal physical times at the same value of n. The purpose of this experiment is to show per-update contraction and finite-grid flattening. If one wants to compare at equal advective time, the relevant nondimensional time is $n\lambda/N$. Record

$$E_2(n) = \|u^n - \bar{u}^n \mathbf{1}\|_2,$$
$$E_\infty(n) = \|u^n - \bar{u}^n \mathbf{1}\|_\infty,$$

$$\mathrm{TV}(n) = \sum_i |u_i^n - u_{i-1}^n|,$$

and

$$M(n) = |\bar{u}^n - \bar{u}^0|.$$

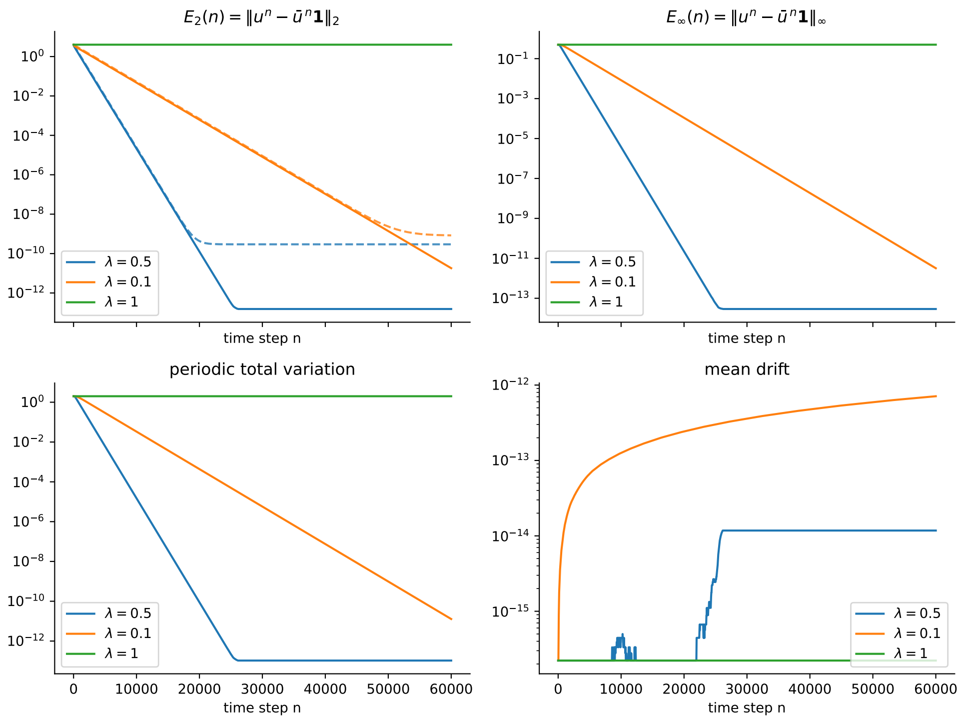

Fig. 4: Periodic step-function flattening for $N = 64$ fine cells and $\lambda \in \{0.5, 0.1, 1\}$. The initial condition is 1 on the first half of the periodic domain and 0 on the second half, so the conserved mean is 1/2. Top left: the L^2 deviation $E_2(n) = \|u^n - \bar{u}^n\mathbf{1}\|_2$; dashed curves show the theoretical bounded-perturbation envelope for $0 < \lambda < 1$, using $\eta = 200\epsilon_{\mathrm{mach}}\sqrt{N}$. Top right: the maximum-norm deviation $E_\infty(n) = \|u^n - \bar{u}^n\mathbf{1}\|_\infty$. Bottom left: the periodic total variation $\mathrm{TV}(n) = \sum_i |u_i^n - u_{i-1}^n|$. Bottom right: the mean drift $M(n) = |\bar{u}^n - \bar{u}^0|$, plotted on a logarithmic scale with a small positive floor for display. For $0 < \lambda < 1$, the nonconstant component decays to the declared perturbation floor. For $\lambda = 1$, the method is a cyclic shift and the step does not flatten.

The results are presented in Figure 4. This is the simplest finite-volume illustration of dissipative decay. The step initially contains sharp unresolved structure. First-order upwind smears that structure for $0 < \lambda < 1$, and the nonconstant component decays about the computed mean until it reaches the perturbation floor. The dashed curves in the E_2 panel use

$$\eta = 200\epsilon_{\mathrm{mach}}\sqrt{N}.$$

This value is a conservative diagnostic perturbation floor used to visualize the bounded-perturbation estimate. It is not derived as a sharp implementation-specific IEEE 754 floating-point rounding constant. The $\lambda = 1$ run is important because it separates advection from dissipation: in that case the method is a cyclic shift and no flattening occurs.

The values in Table 2 quantify the visual message of Figure 4. The nondissipative case $\lambda = 1$ preserves the step by cyclic transport: the final E_2, E_∞, and total variation are unchanged from their initial values. For $\lambda = 0.5$, the nonconstant component is reduced to roundoff-level values by $n = 60{,}000$. For $\lambda = 0.1$, the decay per update is slower, as predicted by the larger value of ρ_N, but the same flattening mechanism is observed. In all cases the measured mean drift remains at roundoff level, so the observed flattening is about a computed mean that remains very close to the initial mean in these runs.

Table 2: Final diagnostics for the periodic step-function experiment at $n = 60{,}000$. The dissipative cases $0 < \lambda < 1$ approach a finite-precision floor, while the control case $\lambda = 1$ preserves the nonconstant profile by cyclic transport.

λ	Final E_2	Final E_∞	Final TV	Max mean drift
1	4.000×10^{0}	5.000×10^{-1}	2.000×10^{0}	0
0.5	1.537×10^{-13}	2.909×10^{-14}	1.048×10^{-13}	1.177×10^{-14}
0.1	1.815×10^{-11}	3.206×10^{-12}	1.282×10^{-11}	7.110×10^{-13}

7.6 Experiment 5: delayed collision and erasure

This experiment addresses the main closure question directly. Take $P = 8$, $r = 8$, $N = 64$, $\lambda = 0.5$, and construct two states x^0 and y^0 with the same parent averages. We use $m = 3$, $\ell = r - 1 - m = 4$, and perturbation amplitude $\delta_0 = 0.25$. The reference state is $y_i^0 \equiv 0.5$. The perturbed state x^0 equals y^0 except in parent $K = 0$, where

$$x^0_{0,0} = 0.5 - \delta_0, \qquad x^0_{0,\ell} = 0.5 + \delta_0.$$

All other child values are 0.5. Hence $Rx^0 = Ry^0$. The compensated perturbation remains inside the same parent through times $0, \ldots, m$, so the parent histories agree initially. The first possible parent-level separation occurs at time $m + 1 = 4$, when part of the perturbation has reached the parent interface.

Measure

$$D(n) = \|RA_\lambda^n x^0 - RA_\lambda^n y^0\|_2.$$

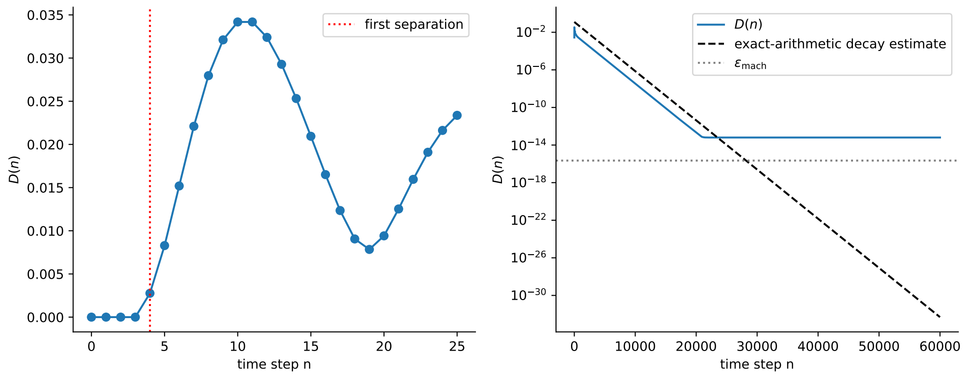

Fig. 5: Delayed collision and dissipative decay for $P = 8$, $r = 8$, $N = 64$, and $\lambda = 0.5$. The two initial fine states have identical parent averages, $Rx^0 = Ry^0$, but differ inside one parent by a compensated subcell perturbation. The plotted quantity is $D(n) = \|RA_\lambda^n x^0 - RA_\lambda^n y^0\|_2$, the difference between the two parent-average states at time step n. Left: short-time behavior. The red dotted line marks the first possible separation time $n = m + 1 = 4$. Before this time, the hidden perturbation has not yet reached a parent interface, so the parent averages agree. After this time, the hidden subcell information becomes visible at the coarse level. Right: long-time behavior on a logarithmic scale. The blue curve decays because the upwind scheme is dissipative. The dashed curve is the exact-arithmetic decay estimate $r^{-1/2}\rho_N^n\|x^0 - y^0\|_2$. Once this estimate falls below the accumulated roundoff scale, the computed curve reaches an implementation-dependent numerical plateau. The gray dotted line marks ϵ_{mach} only as a unit-roundoff reference; it is not expected to coincide with the observed plateau.

Figure 5 combines the two main mechanisms in the paper. The left panel shows that mean-only closure is not exact: two fine states with identical parent averages can produce different future parent averages after a finite delay. The right panel shows that exact memory and practical long-time memory are different. Once the hidden perturbation becomes visible at the parent level, the dissipative upwind scheme contracts it. In exact arithmetic the decay estimate continues toward zero, while in double precision the computed difference reaches an implementation-dependent roundoff plateau. Thus the same perturbation is first hidden, then observable, and finally reduced to the numerical floor of the computation.

7.7 Verification check: one-step flux closure

A final sanity check uses random scalar fine data and verifies the identity in Proposition 1. For the scalar upwind scheme,

$$\Phi_K = \lambda x_{K,r-1}$$

is the realized right-face flux of parent K. The parent update computed by restricting the fine solution is compared with

$$y_K^{n+1} = y_K^n - \frac{1}{r}(\Phi_K - \Phi_{K-1}).$$

The check uses $P = 10$, $r = 4$, $N = 40$, and $\lambda = 0.37$. The fine states are drawn independently from the standard normal distribution using NumPy's default random-number generator with seed 12345. For each of 100 trials, the discrepancy is measured in the maximum norm over parent averages:

$$\left\| RA_\lambda x - \left(Rx - \frac{1}{r}(\Phi - \mathrm{roll}(\Phi, 1)) \right) \right\|_\infty .$$

The maximum discrepancy over the 100 trials was 4.44×10^{-16} (twice ϵ_{mach}), well below 10^{-12} and consistent with roundoff. This check is included only as software validation of the telescoping identity in Proposition 1. The substantive numerical evidence is provided by the rank, conditioning, step-flattening, and delayed-collision experiments.

Taken together, the experiments support the three main claims of the paper. The rank and singular-value experiments compare the numerical construction against the exact memory formula and show where effective rank can differ from exact rank. The queue-conditioning experiment identifies the local temporal mechanism responsible for loss of stable observability, while Proposition 2 records the separate spatial conditioning introduced by the periodic difference operator. The step-function and delayed-collision experiments then show the finite-volume consequences: unresolved information may be needed for exact short-time pathwise closure, but for $0 < \lambda < 1$ the same information can be dissipated below a declared numerical tolerance.

8 Discussion

The results clarify a distinction that is easy to miss in coarse CFD and super-resolution settings. Conservation at one time is not the same as pathwise predictive closure. A reconstruction may exactly preserve parent averages, yet place material differently inside a parent. In an upwind finite-volume method, those placements determine when unresolved data reach parent interfaces. Exact future parent means can therefore require memory beyond the current parent means.

The numerical results in Section 7 are meant to be read in the same order as the analysis. Figure 1 and Table 1 show the gap between exact rank and tolerance-dependent effective rank. Figure 2 shows this gap at the singular-value level, while Figure 3 identifies the triangular delayed-queue map as the local source of poor conditioning. Figure 4 and Table 2 show the dissipative decay predicted by Theorem 3. Finally, Figure 5 combines the two themes: hidden subcell information can affect future parent means after a delay, yet its effect can subsequently decay because of numerical dissipation.

The rank formula gives the exact algebraic requirement for the scalar periodic upwind hierarchy. Each newly exposed child layer contributes $P - 1$ independent periodic differences. The missing one dimension at each level is the constant periodic flux gauge: adding the same face-flux value everywhere does not change any parent average. At saturation, a centralized encoder saves exactly $r - 1$ coordinates compared with raw local queues.

The conditioning theorem gives a different message. Although every positive λ exposes the same algebraic directions, the deepest exposed layer can appear with coefficient λ^q. For small λ, a direction may be visible in exact arithmetic and useless in double precision. This is why the effective rank experiment is necessary. A numerical paper should not report exact dimension without discussing conditioning.

The bounded-perturbation decay theorem gives the final piece. For $0 < \lambda < 1$, the upwind method is diffusive. Nonconstant fine-grid information contracts at rate ρ_N, up to the declared perturbation floor. Thus exact short-time memory and long-time practical memory can differ. The step-function experiment shows this visibly: the sharp initial discontinuity is flattened about the computed mean, whose drift is measured to remain at roundoff level in the reported runs. The delayed-collision experiment combines both effects: two fine states with equal parent averages eventually separate at the coarse level, then their difference is dissipated.

These results do not contradict the value of learned closures, super-resolution, or high-order CFD. They say that a meaningful claim about coarse prediction must specify the information available to the model, the prediction horizon, the tolerance, the numerical method, and the error criterion. Recent CFD and AI studies use different information sets and different goals, including reconstruction quality, short-time forecasting, statistical fidelity, entropy stability, or engineering prediction [28–30]. The present theorem concerns exact pathwise parent-mean prediction in one scalar finite-volume hierarchy.

8.1 Limitations and future work

The analysis is intentionally narrow. The grid is uniform and periodic, the dynamics are linear, the time integrator is forward Euler, and the numerical flux is first-order upwind. The encoder lower bounds are topological dimension statements for continuous real-valued encoders on open sets. They are not bit-complexity bounds, statistical learning bounds, or claims about arbitrary discontinuous encoders. The results also do not establish the same rank formula for nonlinear conservation laws, compressible Euler dynamics, nonuniform or unstructured grids, nonperiodic boundaries, high-order reconstruction, Runge–Kutta stages, limiters, entropy constraints, stochastic closures, or learned models.

These limitations are useful because they identify the next problems. A natural extension is to repeat the same program for higher-order scalar schemes: identify the exact observable subspace, quantify its conditioning, and test which parts are preserved or erased by numerical dissipation. Another direction is to study nonperiodic boundaries, where the periodic flux gauge is replaced by boundary-dependent information. A carefully scoped Euler study should begin with the one-step flux identity of Proposition 1, then ask whether a finite predictive register can be defined for a declared solver, limiter, and time integrator. More complex finite-volume closures may involve flux histories, interface states, residual stresses, or memory terms [14, 15, 18]. The same three questions should guide such extensions: what is the exact predictive state, how well conditioned is it, and which parts of it survive numerical dissipation?

9 Conclusions

Coarse finite-volume averages are generally not exact predictive states for the coarse evolution of a realized fine solution. In the periodic scalar upwind hierarchy studied here, the exact finite-horizon memory requirement is

$$(P-1)\min\{L, r-1\}$$

real coordinates beyond the current parent averages. A flux-divergence queue attains this bound, and periodicity removes one common flux-gauge coordinate per queue time.

The main point is not only the rank formula. Exact rank, stable recovery, and long-time numerical relevance are distinct. Delayed collar information enters the queue with powers of λ, so algebraically visible modes can fall below an SVD tolerance when λ is small. For $0 < \lambda < 1$, the same upwind method dissipates nonconstant modes, so unresolved subcell information that matters for exact short-time closure may later become irrelevant below a declared perturbation floor.

The numerical experiments separate these effects. The rank and singular value tests reproduce the predicted rank ladder and display the effective-rank loss. The queue-conditioning test explains the loss through the triangular collar-to-queue map. The step-function test shows flattening about the computed mean for $0 < \lambda < 1$, while the $\lambda = 1$ control confirms that advection without numerical diffusion does not erase the step. The delayed-collision test shows the closure mechanism directly: same-parent-mean states can separate at the coarse level after hidden subcell data reaches an interface, and the resulting difference is then damped.

The practical lesson is that a coarse model should not be judged only by whether it preserves parent averages at one time. One must also specify the hidden information available to the model, the prediction horizon, the floating-point tolerance, the numerical method, and whether the method preserves or erases unresolved information over that horizon. This paper establishes that perspective in a scalar periodic model; the limitations and extensions identified in Section 8.1 describe the natural next steps.

Declarations

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Author contributions

A.P. contributed to conceptualization, interpretation, verification, and writing. N.C. contributed to conceptualization, mathematical analysis, numerical analysis, software, interpretation, verification, and writing. D.D. contributed to conceptualization, numerical analysis, interpretation, verification, and writing.

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Data and code availability

The complete reproducibility package for this study is publicly available in the Certified Simulation repository of the University of Nicosia, https://github.com/UniversityOfNicosia/certified-simulation, under `papers/finite-horizon-memory`. The package contains the Python scripts that generate every matrix test, scalar finite-volume experiment, CSV file, LaTeX table fragment, and figure reported in the manuscript, together with instructions for reproducing the results, and executable certificates that replay the paper's exact results in exact rational arithmetic. No external CFD solver or proprietary software is required.

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